

Roof Lifecycle Cost Analysis

Planned maintenance is projected to save the owner of a typical Hampton Roads warehouse **\$913K** over the next 20 years.

About 29% less than reactive repair: \$583K lower spending plus \$330K more roof life at year 20. Range \$434K–\$1.23M from conservative to upside assumptions; \$490K in today's dollars at a 7% discount rate.

MARKET	PROPERTY	ROOF AREA	SYSTEM	AGE · CONDITION	NEW ROOF TODAY
Hampton Roads	Warehouse	100,000 sq ft	TPO	8 yrs · Good	\$900K

PAXSEAL NET COST

\$2.29M

\$2.20M spent + \$540K roof used – \$450K roof left

REACTIVE NET COST

\$3.20M

\$2.78M spent + \$540K roof used – \$120K roof left

VALUE PROTECTED

\$447K

\$34K/yr more NOI at a 7.5% cap rate, if the owner pays for the roof

ROOF PRICES

2× in 15 yrs

At 4.6%/yr, the long-run trend in commercial roofing costs

Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.



Roof life

When each scenario needs a new roof.



Where the savings come from

20-year totals

	PAXSEAL	REACTIVE	SAVINGS
PaxSeal fees	\$371K	—	-\$371K
Repairs paid by owner	\$219K	\$1.29M	+\$1.07M
Expected storm damage	\$131K	\$197K	+\$66K
Roof replacements	\$1.48M	\$1.29M	-\$190K
Cash spending	\$2.20M	\$2.78M	+\$580K
Current roof value used	\$540K	\$540K	—
Roof life left at year 20	-\$450K	-\$120K	+\$330K
Net 20-year cost	\$2.29M	\$3.20M	+\$910K

What this means

- 1 **A later replacement.** The next new roof moves from year 8 to year 11, leaving 10 years of roof life at year 20 instead of 2.
- 2 **Smaller repairs.** Emergency repairs run about 3× the planned rate, and the PaxSeal fee covers half of planned repairs. Together that saves \$1.07M.
- 3 **The program pays for itself.** Each \$1 of fees avoids about \$2.57 of other cash roof costs.

Local risk profile

Norfolk city, VA (51710)

Hurricane / tropical-storm wind

About 4.8 tropical storms or stronger within 50 nautical miles of Norfolk per 20 years (NOAA HURDAT2 1975–2024). Expected storm damage to this roof over 20 years is **\$131K** if maintained and **\$197K** with reactive repair only, before insurance. A documented maintenance history also helps when an insurer reviews a claim.

Hail	Relatively Low	■ ■ ■ ■ ■
Strong wind	Relatively Moderate	■ ■ ■ ■ ■
Tornado	Relatively Moderate	■ ■ ■ ■ ■
Hurricane	Relatively Moderate	■ ■ ■ ■ ■
Winter weather	Relatively Moderate	■ ■ ■ ■ ■
Ice storm	Relatively Moderate	■ ■ ■ ■ ■
FEMA National Risk Index rating		

Key assumptions

The basis for each figure in this analysis.

ASSUMPTION	VALUE	BASIS
Roof life	20 / 15 yrs	Maintained / reactive. Industry life tables; maintenance adds ~5 yrs (NRC Canada, warranties)
Roof cost growth	4.6%/yr	BLS Producer Price Index, nonresidential roofing, 2007–2025
Replacement cost	\$9.00/sq ft	Hampton Roads contractor price ranges
Reactive repair cost	3× planned	Industry estimates. PaxSeal fee covers half of planned repairs
Storm damage	0.44%/yr	Of roof cost, before insurance; ×1.5 if reactive. FEMA National Risk Index
PaxSeal fee	\$13,800/yr	Year 1, rising 3%/yr. Estimate; your proposal states the actual fee
Cap rate	7.5%	Hampton Roads warehouse / industrial market reports

Next step: a roof assessment and a written PaxSeal proposal for this building. Call PAX Services Group at 1-855-729-7510 or email info@paxserv.com.

Estimates, not a quote. Projections from Hampton Roads market research and industry data, in nominal dollars. The PaxSeal fee is an estimate; your proposal states the actual price. Net cost includes wear on the current roof, less roof life left at year 20; both roof-value figures are in today's dollars. Storm damage is before insurance. Property value is the yearly gain in net operating income divided by the cap rate, for income property where the owner pays for the roof.